



Elliptic PDEs: Poisson and Laplace Equations

Definition

Let $u(x, y)$ be twice differentiable with $x \in \mathcal{X} = [x_a, x_b]$ and $y \in \mathcal{Y} = [y_a, y_b]$, and define $f(x, y)$ for $(x, y) \in \mathcal{X} \times \mathcal{Y}$. The **Poisson Equation** is given by

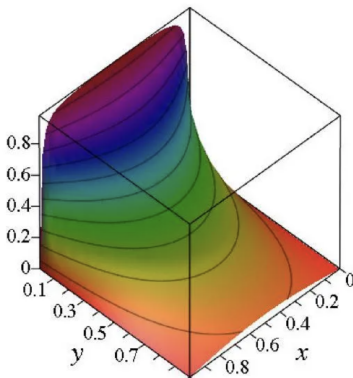
$$u_{xx} + u_{yy} = f(x, y).$$

For the choice $f(x, y) = 0$, the Poisson equation becomes the **Laplace Equation**.

These represent a third type of PDE, elliptic equations, which are time independent.

Elliptic PDEs: Laplace Equation

Example application: Solving for the steady state temperature profile when there are different boundary conditions





Laplace Equation

The 2-dimensional Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ with Dirichlet boundary conditions has four boundary conditions

$$u(x, y_b) = g_1(x), \quad \text{temperature on bottom}$$

$$u(x, y_t) = g_2(x), \quad \text{temperature on top}$$

$$u(x, y_l) = g_3(y), \quad \text{temperature on left}$$

$$u(x, y_r) = g_4(y), \quad \text{temperature on right}$$

Note there is no initial condition since the solution $u(x, y)$ does not depend on time.

Solving with Finite Difference Method

We can approximately solve the 2-dimensional PDE by approximating it with a linear system based on finite differences:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{k^2} + \mathcal{O}(h^2, k^2),$$

where $x_i = x_0 + hi$ is a mesh in the x direction with step size h and $y_j = y_0 + kj$ is a mesh in the y direction with step size y .

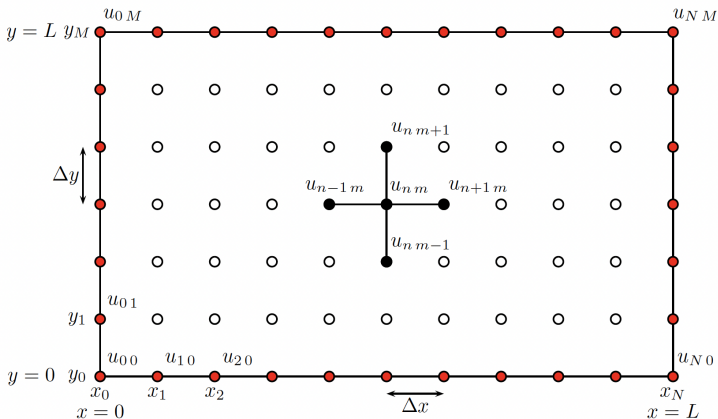
We typically choose $h = k$, yielding a system of $(n-1) \times (n-1)$ equations of the form:

$$u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j} = h^2 f(x_i, x_j)$$

where $i, j = 1, 2, \dots, n-1$.

Mesh for 2D Elliptic PDEs

2D mesh and stencil for Poisson equation.





Solving with Finite Difference Method

The linear equations

$$u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j} = h^2 f(x_i, x_j)$$

for $i, j \in \{1, 2, \dots, n-1\}$ can be rewritten in the form

$$A\hat{\mathbf{u}} = \hat{\mathbf{b}}$$

by stacking the $u_{i,j}$ values into a very long vector $\hat{\mathbf{u}} \in \mathbb{R}^{(n-1)^2}$.

Similarly, $\hat{\mathbf{b}} \in \mathbb{R}^{(n-1)^2}$ encodes the boundary conditions and $h^2 f(x_i, y_j)$ values. Matrix $A \in \mathbb{R}^{(n-1)^2 \times (n-1)^2}$ is very large and sparse.

(See notes.)



Laplace Equation Example

EXAMPLE: 2-dimensional Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ with four Dirichlet boundary conditions

$$u(x, y_b) = g_1(x) = \sin(2\pi x)$$

$$u(x, y_t) = g_2(x) = \sin(2\pi x)$$

$$u(x, y_l) = g_3(y) = 2 \sin(2\pi y)$$

$$u(x, y_r) = g_4(y) = 2 \sin(2\pi y)$$

(See code.)